ELEC 221 Lecture 13
CT Fourier transform: convolution, multiplication; the magnitude-phase representation; step response

Thursday 22 October 2026

Announcements

- TA3 due tomorrow 23:59
- Monday is last tutorial before midterm (midterm prep)
- Midterm exam cutoff is next Thursday's class
- Quiz 7 Tuesday
- Final exam time announced: Friday 18 December 15:30-18:00
- Tomorrow's office hours on Zoom

Last time

We evaluated Fourier spectra for *periodic* functions.

Last time

We observed time-frequency duality in the Fourier transform.

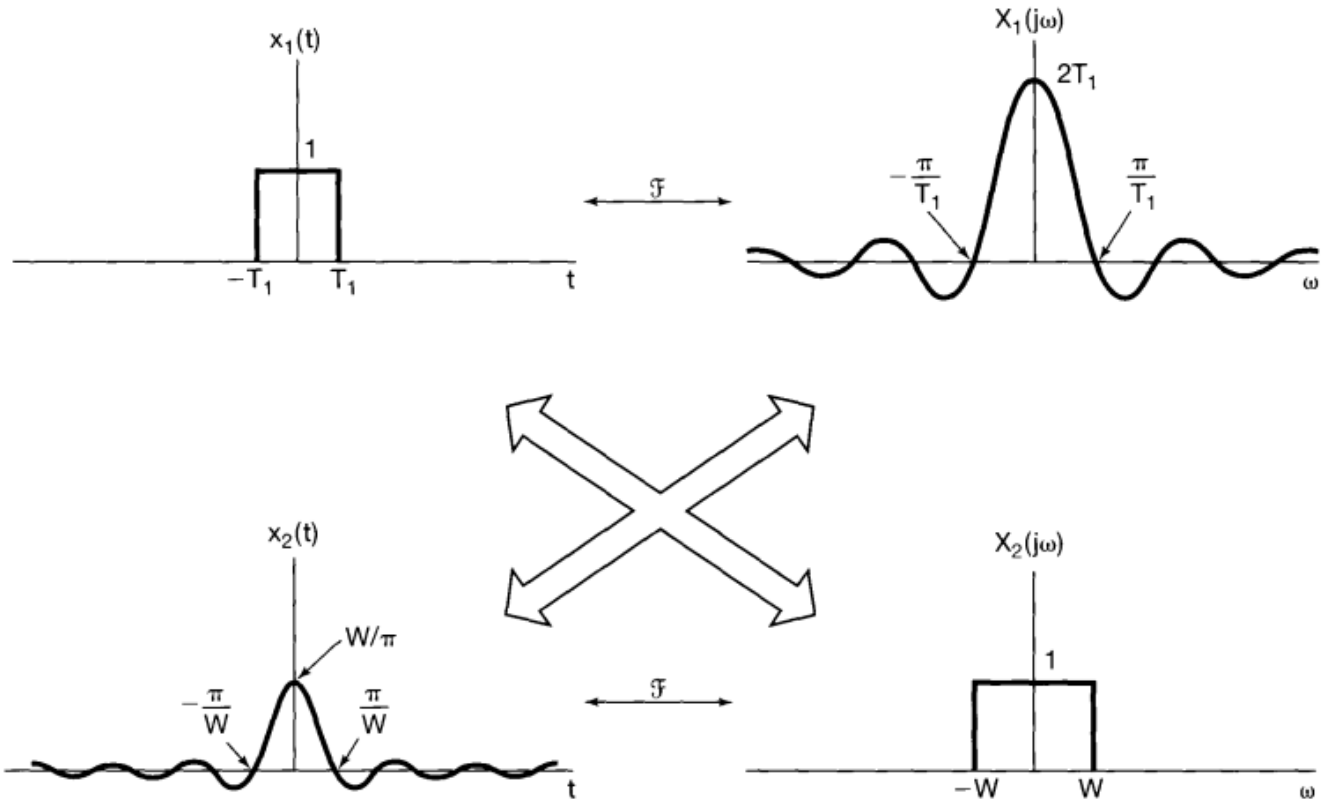


Image credit: Oppenheim chapter 4.3

Last time

We derived a very important property of Fourier transforms under *convolution*:

This makes it easier to analyze LTI systems!

Today

Learning outcomes:

- Apply the *convolution property* of the Fourier transform to characterize LTI system behaviour
- Describe the *multiplication property* of the Fourier transform and provide applications of its use
- Express a Fourier transform using the magnitude-phase representation and read a Bode plot
- Compute the step response of LTI systems described by first-order ODEs

Example: convolution

Suppose $x(t) = \frac{\sin(\omega_0 t)}{\pi t}$ is input into a lowpass filter

Exercise: convolution

Consider an LTI system that sends

What is its impulse response?

Exercise: convolution

Intuition: cascade of systems

What is the frequency response of the following setup?

The multiplication property

We know that:

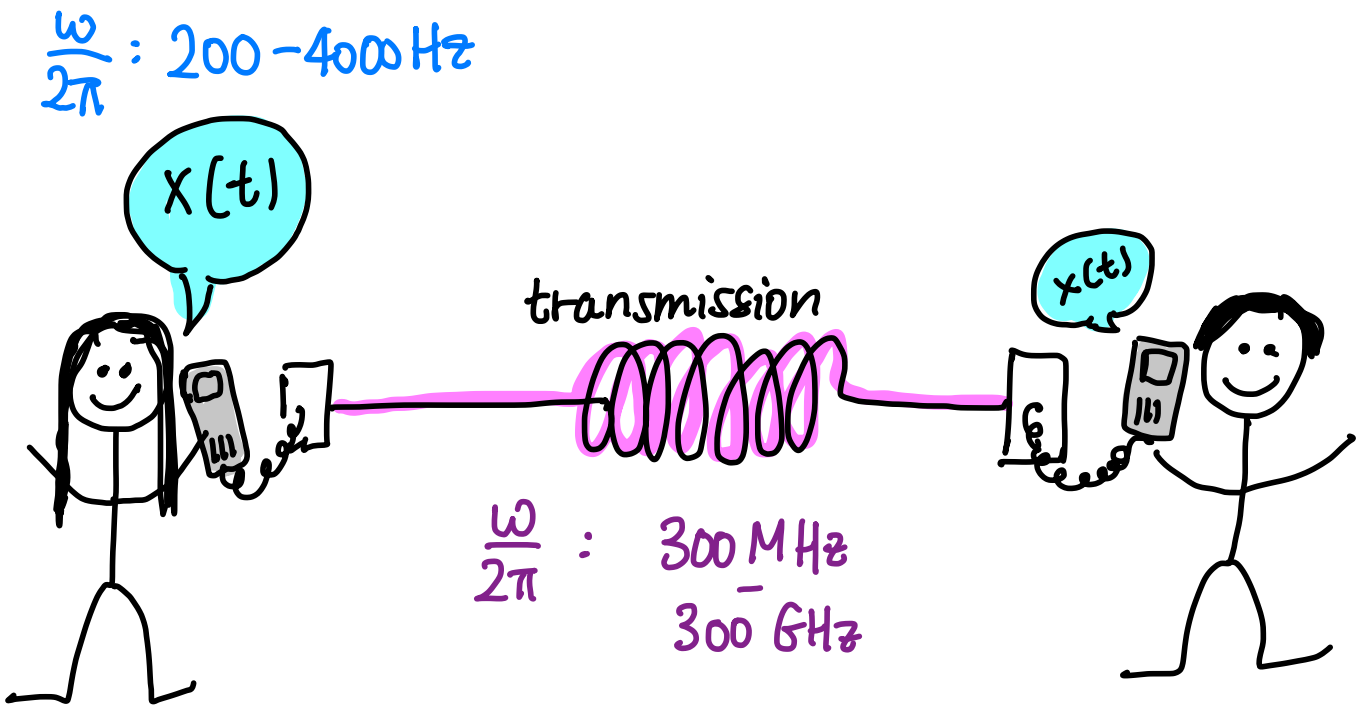
Something similar holds when we interchange time and frequency:

This is the **multiplication property**.

Example: the multiplication property

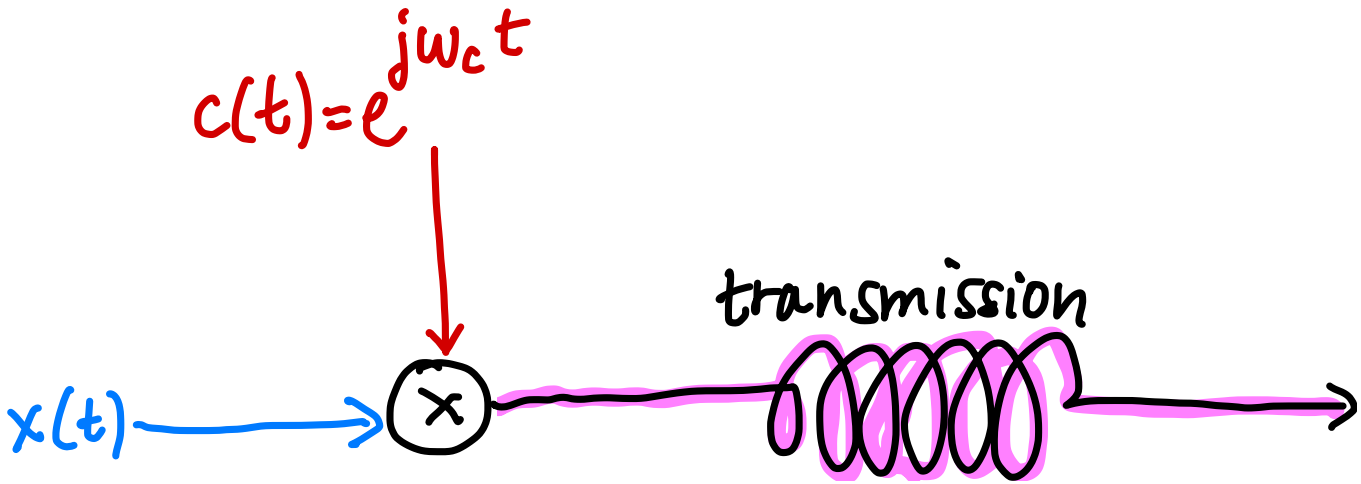
We will explore this more in depth when we discuss communication systems and signal **modulation**.

For now, here is a taste:



Example: the multiplication property

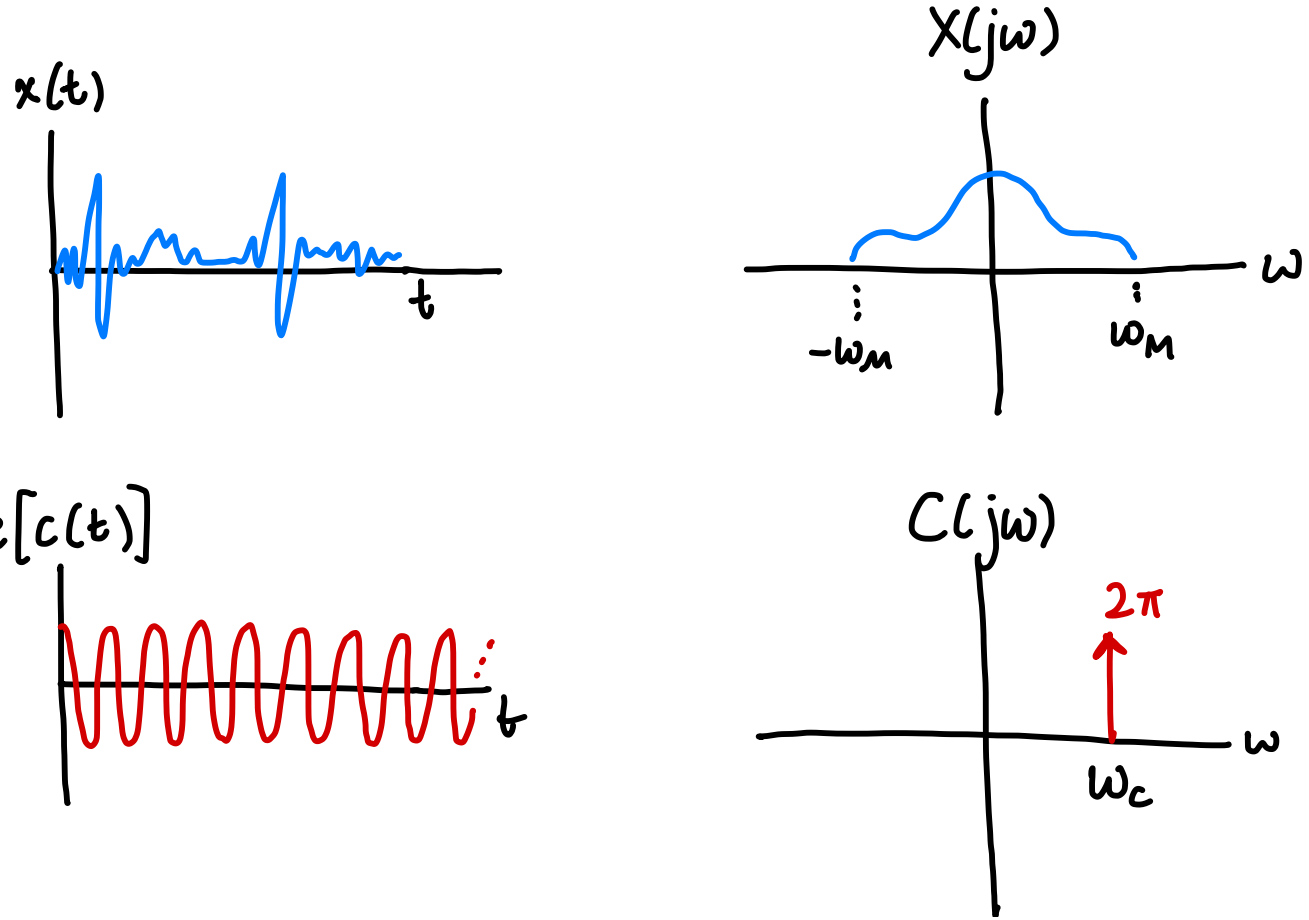
To shift our signal into the frequency range of transmission, we can multiply it by a **carrier signal** (amplitude modulation):



Is this doing what we think it is?

Example: the multiplication property

Consider the Fourier spectrum of both signals:



Example: the multiplication property

The multiplication property tells us

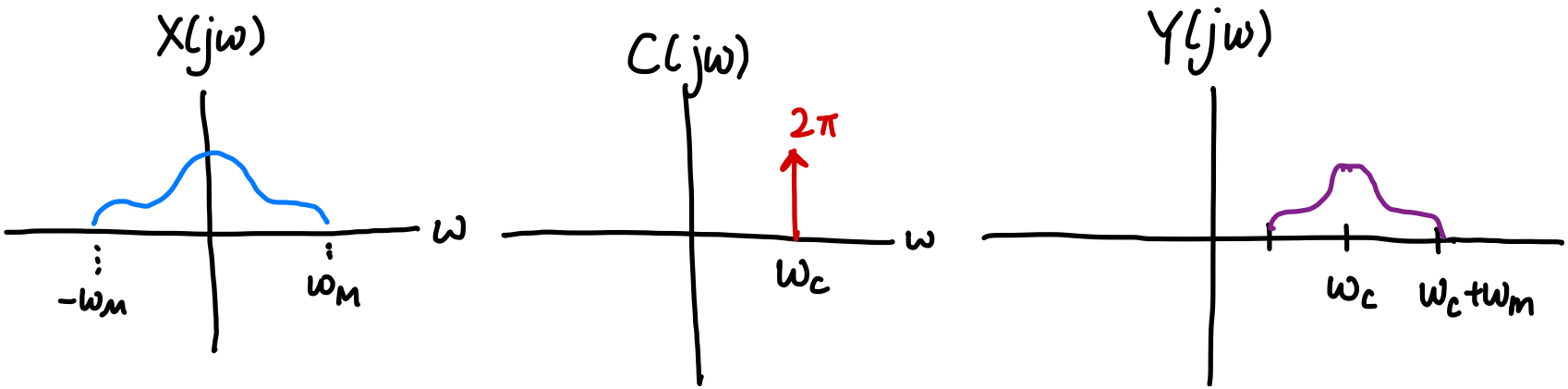
We have

Example: the multiplication property

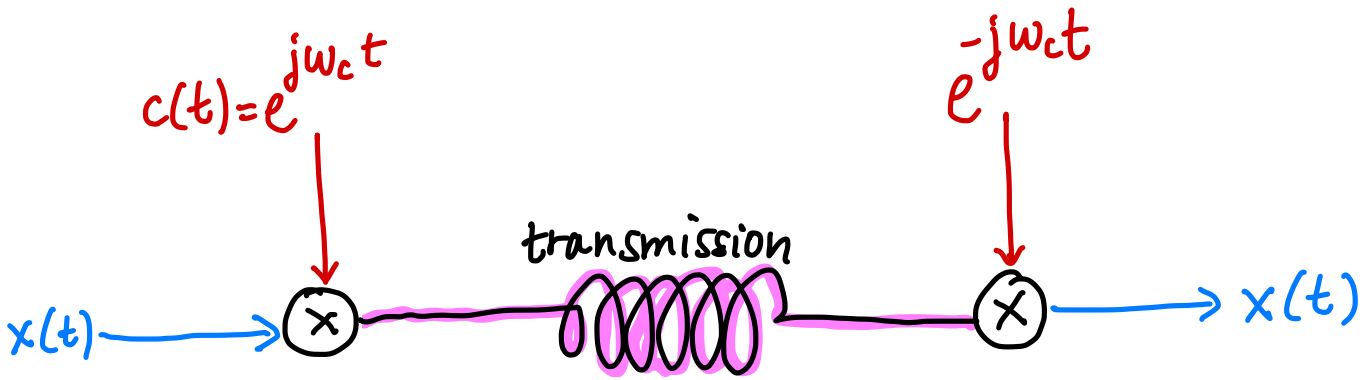
Let's convolve them:

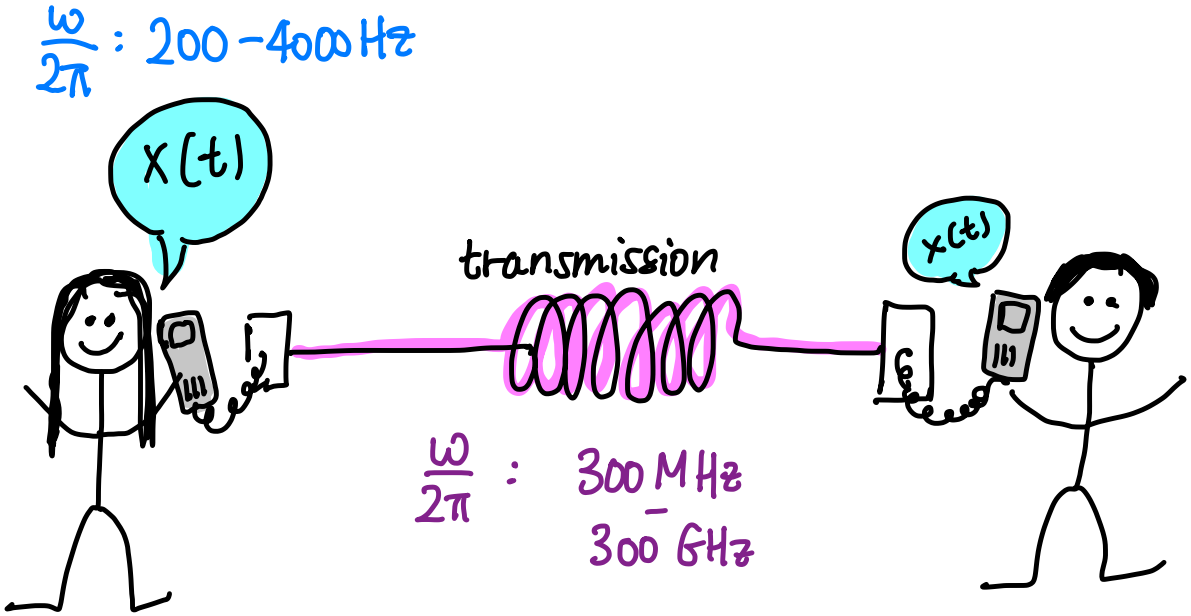
Example: the multiplication property

Multiplication with complex exponential carrier signal shifts the spectrum. We can move it into the desired frequency range.



Example: the multiplication property





Example: frequency-selective filtering with variable centre frequency

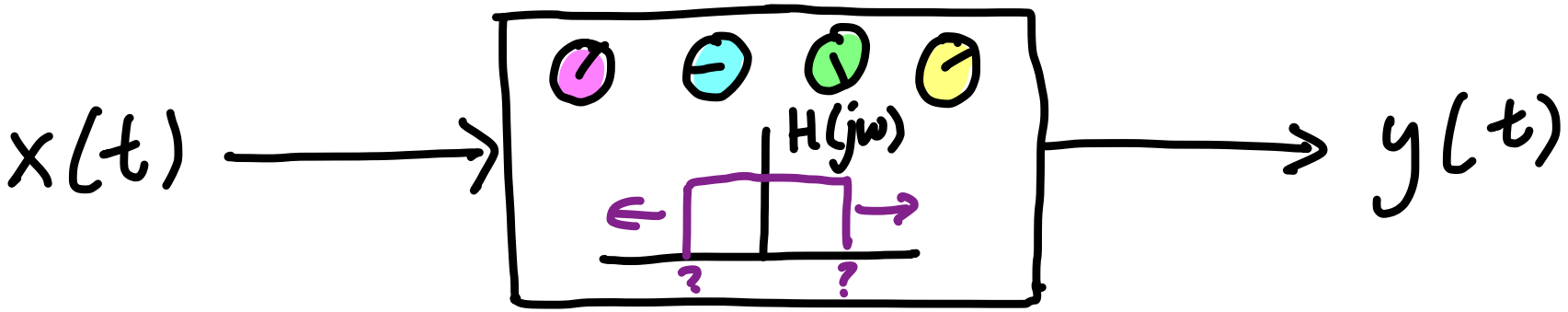
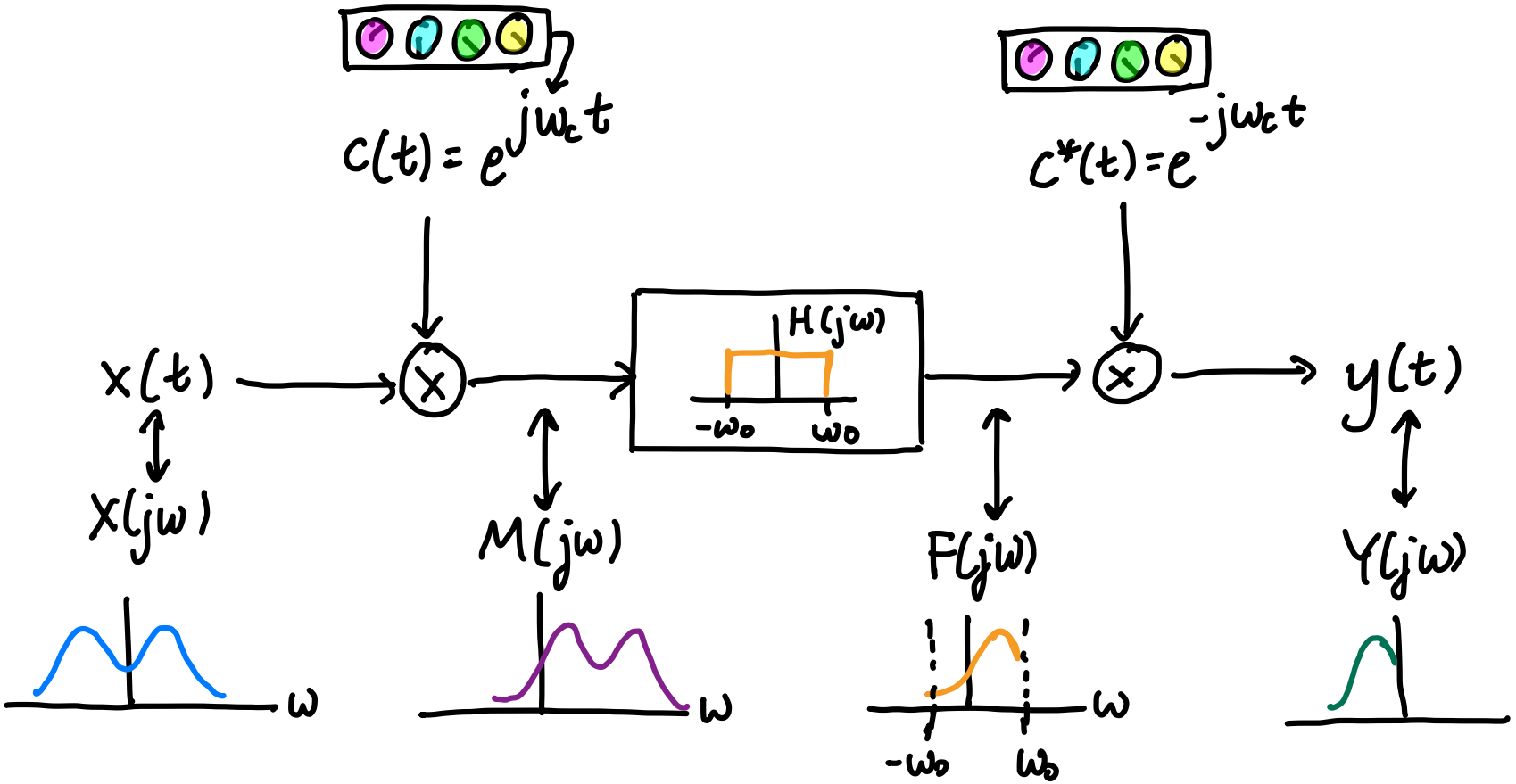




Image: `https://www.euronics.ee/UserFiles/Products/Images/162391-panasonic-rf-2400d-radio.png`

Example: frequency-selective filtering with variable centre frequency



The magnitude-phase representation

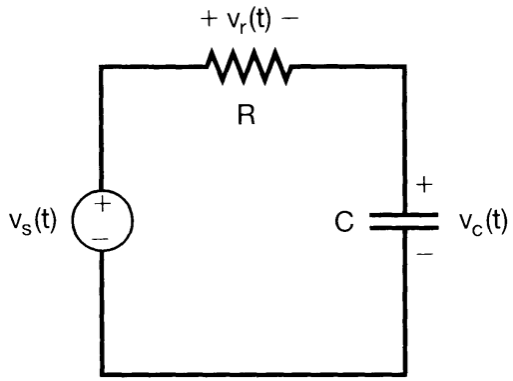
Since Fourier spectra are complex we can express them in terms of their magnitude and phase.

Recall the convolution property:

Exercise: How does a system $H(j\omega)$ affect $|X(j\omega)|$ and $\sphericalangle X(j\omega)$?

Example: Lowpass filters in practice

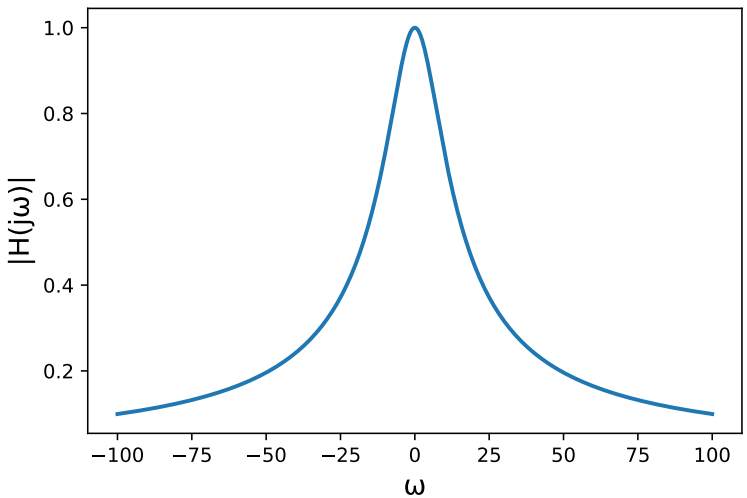
The magnitude-phase representation helps visualize and characterize systems. Recall our RC circuit / lowpass filter.

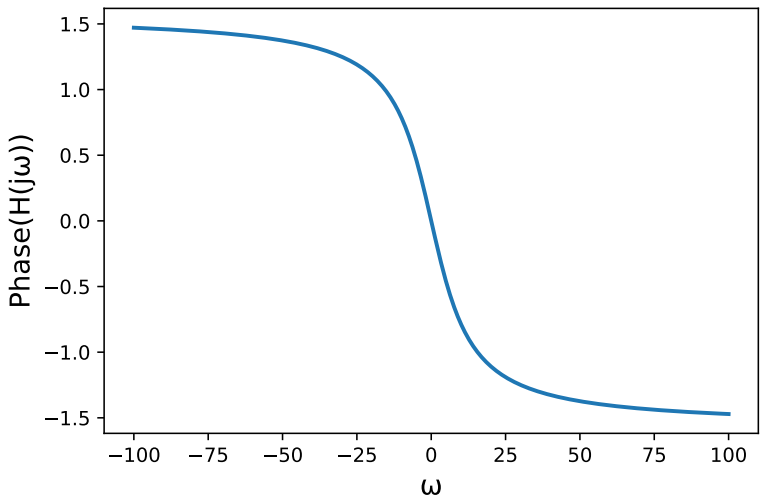


We wrote a differential equation for the system and found

Example: lowpass filters in practice

Results in the following frequency response (setting $RC = 0.1$):





Adjusting RC controls the frequency response; the phase shift is also frequency dependent.

Bode plots

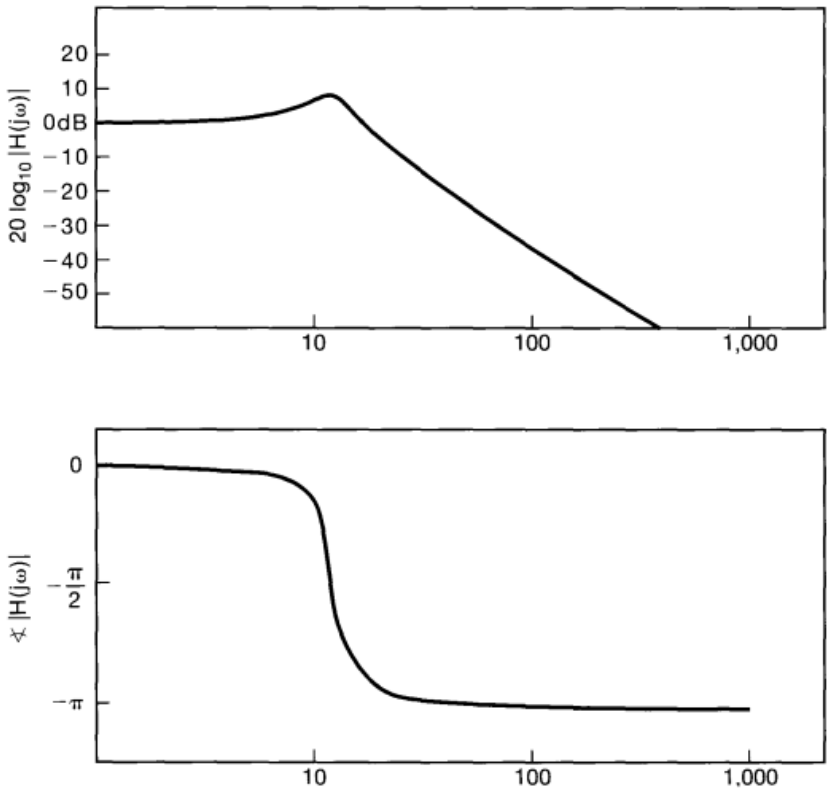
Recall:

Magnitude is multiplicative and phase is additive... would be nicer if both were additive.

Rather than plotting $|H(j\omega)|$ and $\sphericalangle H(j\omega)$, it is common to plot $20 \log_{10} |H(j\omega)|$ and $\sphericalangle H(j\omega)$ against $\log_{10} \omega$.

Bode plots

These are called *Bode plots*:



The logarithmic scale also allows us to view the response over a much wider range of frequencies.

Image credit: Oppenheim 6.2

Bode plots: first-order systems

$$T\frac{dy(t)}{dt} + y(t) = x(t), \qquad H(j\omega) = \frac{1}{1 + j\omega T}$$

Let's view these in the magnitude-phase representation:

From this, we find

Bode plots: first-order systems

We have

To make our Bode plot, compute

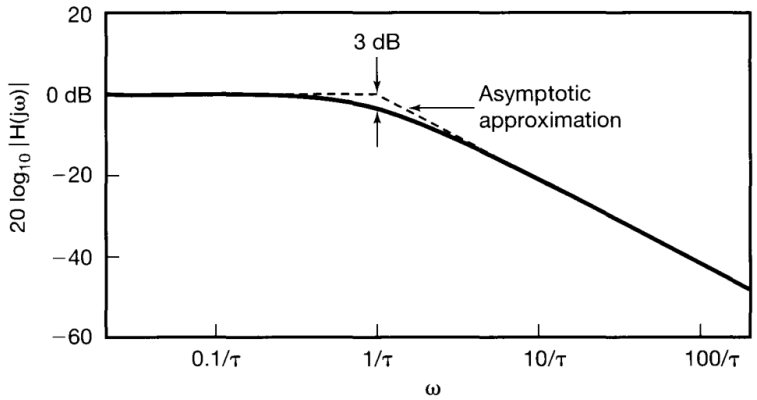
Bode plots: first-order systems

If $\omega << 1/T$,

If $\omega >> 1/T$, $\omega T >> 1$

If $\omega = 1/T$,

Bode plots: first-order systems



Can make similar approximations to recover plot of phase

$$H(j\omega) = \tan^{-1}(-\omega T)$$

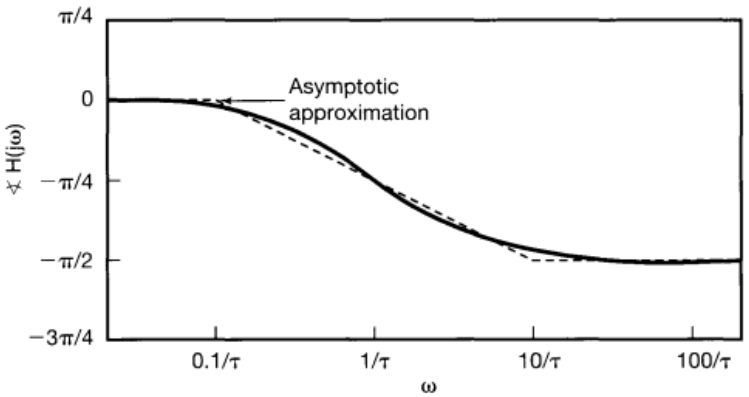
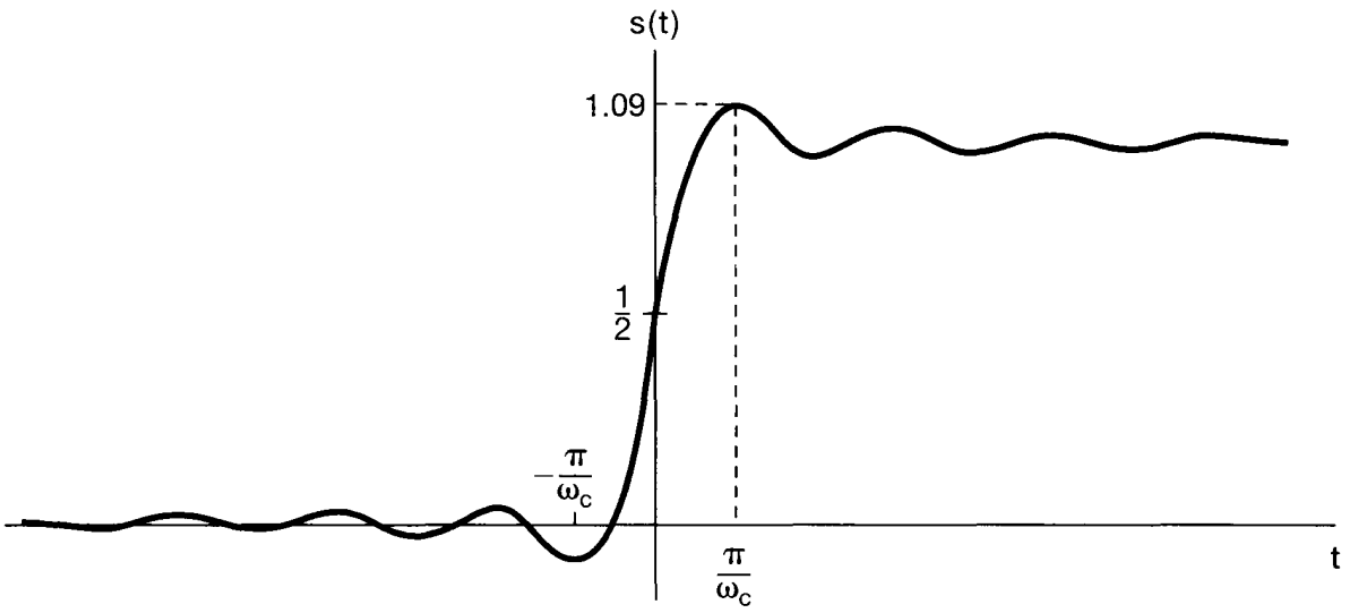


Image credit: Oppenheim 6.5

Step response

It is also important to consider the *step response* of filters:

Ideal filter step response



An ideal filter leads to **ringing** in the step response.

Image credit: Oppenheim 6.3

Ideal filter step response

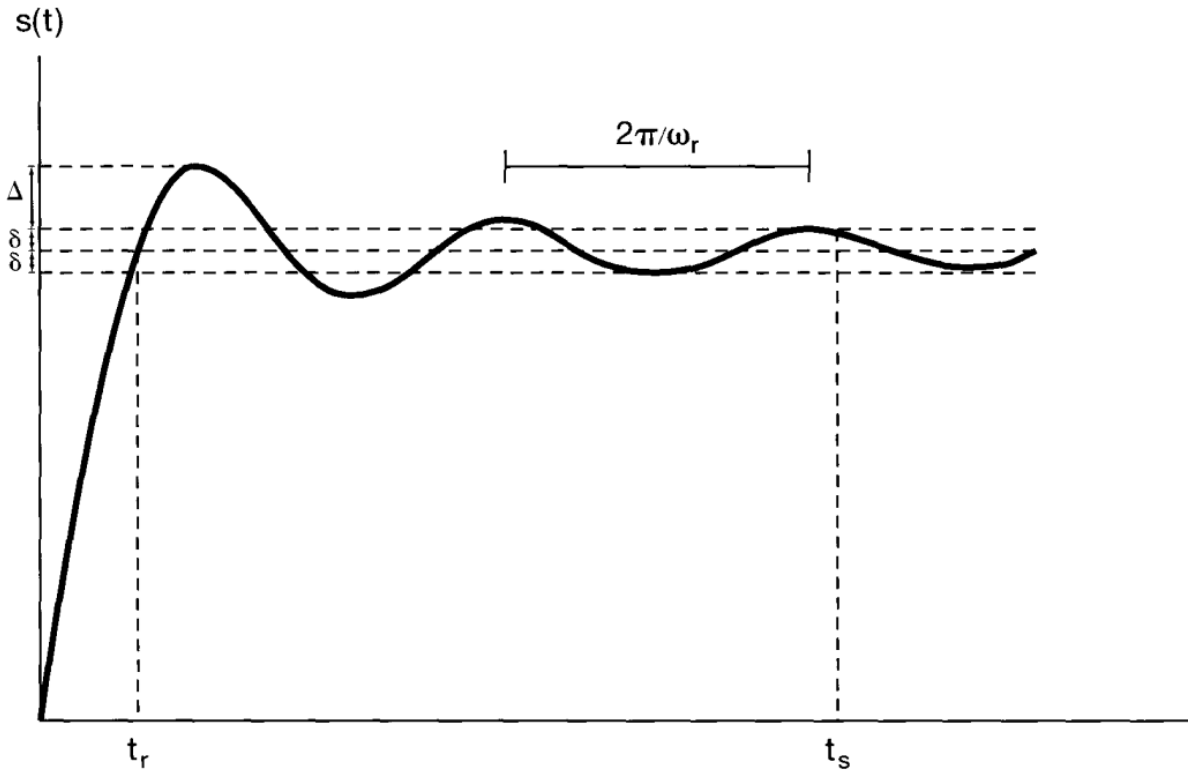


Figure 6.17 Step response of a continuous-time lowpass filter, indicating the rise time t_r, overshoot Δ, ringing frequency ω_r, and settling time t_s—i.e., the time at which the step response settles to within $\pm\delta$ of its final value.

Image credit: Oppenheim 6.4

Non-ideal filters

There are **tradeoffs** in filter design. Compromises in the frequency domain can lead to nicer behaviour in the time domain.

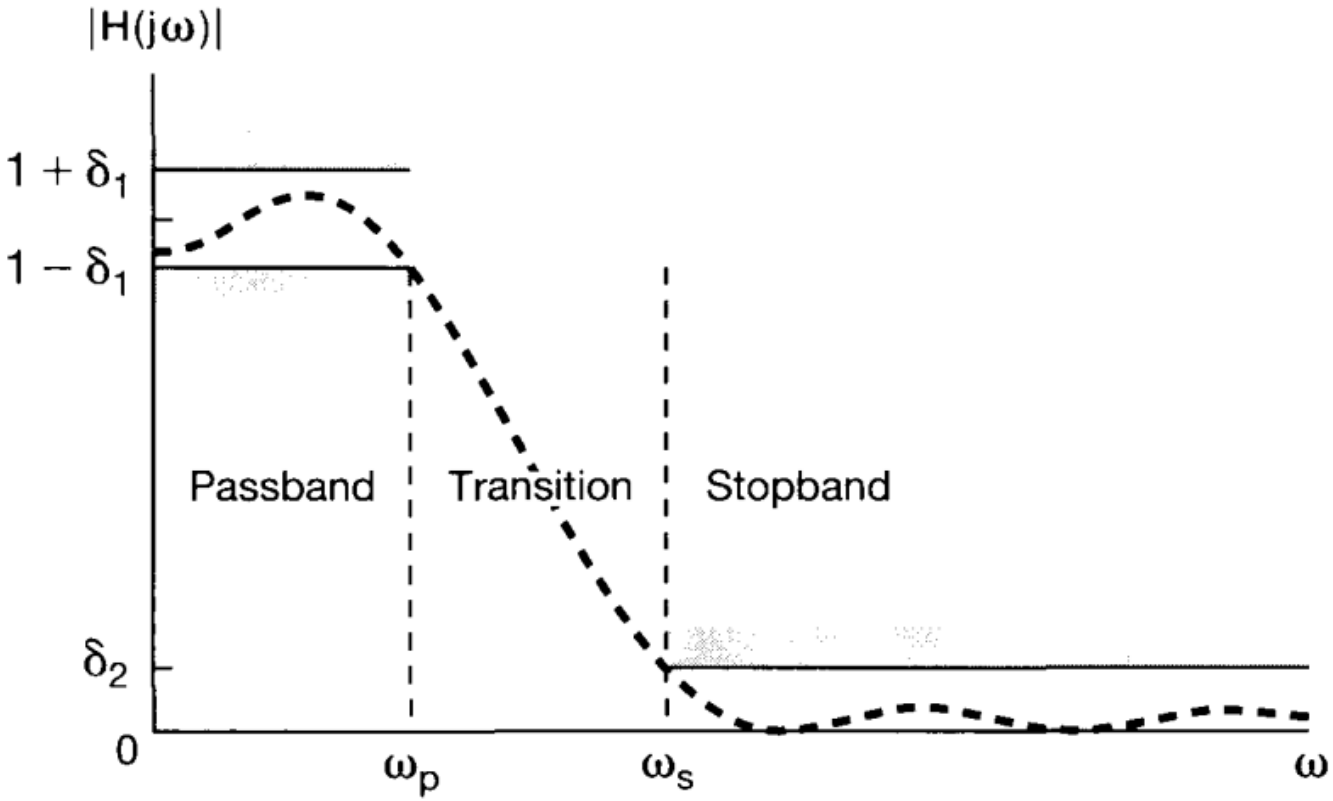


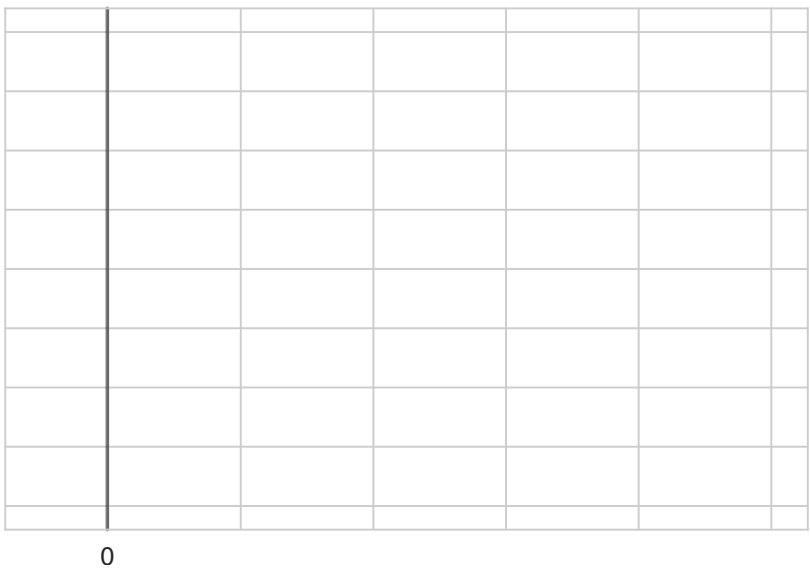
Image credit: Oppenheim 6.4

Exercise: step response of first-order systems

Exercise: step response of first-order systems

The impulse and step response of the system are

T is the **time constant** of the system.



For next time

Content:

- CT Fourier transform under differentiation and integration
- Systems described by second-order ODEs

Action items:

1. Tutorial assignment 3
2. Prepare for Quiz 7
3. Study for midterm

Recommended reading:

- For today's class: Oppenheim 4.3-4.7, 6.1-6.2
- Suggested problems: 4.5, 4.9, 4.11-4.16, 4.19, 4.22, 4.25, 4.26, 4.29, 6.1, 6.3a, 6.5, 6.9
- For next class: Oppenheim 4.7, 6.3-6.5